

Donghwi Jung
Samsung Electronics
Future Robotics Office - Humanoid Team
Ph.D., Seoul National University, 2026
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Research Interests

Robotics, SLAM, Mapping, Localization, Vision-Language Navigation, Human-Robot Collaboration, Bio-inspired Grounding for Embodied Intelligence

Education

Seoul National University Sep. 2020 - Feb. 2026
Ph.D. in Civil & Environmental Engineering
Advisor: Prof. Seong-Woo Kim
Dissertation: “Spatial and Linguistic Understanding for Autonomous Navigation in Smart Cities”

Seoul National University Mar. 2008 - Feb. 2017
B.S. in Food Science & Biotechnology / B.A. in Business Administration

Publications

Kyung-Rok Rho, Jooyoung Bae, Mintaek Oh, **Donghwi Jung**, Seung-Woo Seo, and Seong-Woo Kim, “*CarryMimic: Stable Whole-Body Humanoid Object Carrying via Force-Closure Reward*,” submitted to Conference on Robot Learning (CoRL 2026)

Mintaek Oh, Kyung-Rok Rho, Hanbi Baek, Jooyoung Bae, Jung-Taak Kim, **Donghwi Jung**, Seung-Woo Seo, and Seong-Woo Kim, “*Embodied Agentic Memory: Vocabulary-Free Entity-Event Memory for Long-Term Agents*,” submitted to Conference on Robot Learning (CoRL 2026)

Jinhee Kim, Dong-Yeop Shin, Stanislaw Sommerfeld, Dongmin Shin, **Donghwi Jung**, and Seong-Woo Kim, “*Language as Control: A Unified Information-Theoretic Model of Trust, Cooperation, and Antagonism*,” submitted to NeurIPS 2026.

Donghwi Jung, Jihun Moon, Jiyang Lee, DongYeop Shin, Jinhee Kim, Mingyu Kim, Jungjae Lee, Seong-Woo Kim, “*Beyond Destinations: Instruction-Aware Graph Path Planning and Navigation with OpenStreetMap*,” IEEE Robotics and Automation Letters (RA-L), 2026. (under review)

Donghwi Jung, Keonwoo Kim, Seong-Woo Kim, “*GOTPR: General Outdoor Text-based Place Recognition Using Scene Graph Retrieval with OpenStreetMap*,” IEEE Robotics and Automation Letters (RA-L), May 2025.

Chan Kim*, Keonwoo Kim*, Mintaek Oh, Hanbi Baek, Jiyang Lee, **Donghwi Jung**, Soojin Woo, Younkyung Woo, John Tucker, Roya Firoozi, Seung-Woo Seo, Mac Schwager, and Seong-Woo Kim, “*E2Map: Experience-and-Emotion Map for Self-Reflective Robot Navigation with Language Models*,” 2025 IEEE International Conference on Robotics and Automation (ICRA), Atlanta, USA, May 2025.

Donghwi Jung, Andres Pulido, Jane Shin, Seong-Woo Kim, “*Point Cloud Structural Similarity-Based Underwater Sonar Loop Detection*,” IEEE Robotics and Automation Letters (RA-L), March 2025.

Projects

<i>Perceptive Humanoid Locomotion Control</i> Internal Project Samsung Electronics Future Robotics Office Role – Methodology Development, Environment Setup, Model Training, and Experimental Validation	Mar. 2026 – Present
<i>Visual-Language Outdoor Navigation Using OpenStreetMap</i> Internal Laboratory Project Role - Navigation Algorithm Implementation, Data Acquisition and Experimentation, Team Leadership	Apr. 2025 - Sep. 2025
<i>Visual-Language Indoor Navigation Using Foundation Models</i> Internal Laboratory Project Role - 2D/3D Mapping, Virtual Environment Implementation, Visual-Language Map Generation, and Localization	Apr. 2024 - Sep. 2024
<i>Development of strong autonomous intelligence for unmanned vehicles</i> R&D Project funds from the Agency for Defense Development, Republic of Korea Role - LiDAR Odometry Estimation	Jan. 2022 - July 2023
<i>Location estimation method in pretreatment and painting factory using inertial sensors</i> Industrial Cooperative Project funds from Daewoo Shipbuilding & Marine Engineering Co., Ltd Role - Inertial Odometry Estimation	Sep. 2022 - Feb. 2023
<i>Yard 3D map creation and virtual environment establishment</i> Industrial Cooperative Project funds from Daewoo Shipbuilding & Marine Engineering Co., Ltd Role - Data Acquisition, Virtual Environment Implementation, 3D Mapping	Apr. 2022 - Oct. 2022
<i>Development of AI-based transporter safety driving technology</i> Industrial Cooperative Project funds from Daewoo Shipbuilding & Marine Engineering Co., Ltd Role - Data Acquisition, Mapping, and Object Detection	Sep. 2021 - Sep. 2022

Teaching Experience

<i>Teaching Assistant</i> How to Make a Robot with Artificial Intelligence College of Engineering, Seoul National University	Fall. 2021 - Spring. 2022
<i>Teaching Assistant</i> Basic Computing: First Adventures in Computing Faculty of Liberal Education, Seoul National University	Summer. 2021 - Spring. 2022
<i>Teaching Assistant</i> Basic Mathematics and Programming Practice for Machine Learning College of Engineering, Seoul National University	Spring. 2021

Global Experience

<i>Visiting Student</i> Active Perception and Robot Intelligence Lab, University of Florida Advisor: Prof. Jane Shin Research on sonar-based underwater image super-resolution and point cloud loop detection	Jul. 2023 – Feb. 2024
<i>Exchange Student</i> University of Singapore	Jul. 2013 – May. 2014

Presentation

<i>IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)</i> Hangzhou, China Title – GOTPR: General Outdoor Text-based Place Recognition Using Scene Graph Retrieval with OpenStreetMap	Oct. 22, 2025
<i>IEEE International Conference on Robotics and Automation (ICRA)</i> Atlanta, USA Title – E2Map: Experience and Emotion Map for Self-Reflective Robot Navigation with Language Models	May 22, 2025
<i>The 18th KICS Future Communication Technology Workshop</i> Seoul, Republic of Korea Title – Language Understanding for Spatial Understanding: Human-Robot Communication	May 23, 2025

Awards

<i>Best Paper Award (Bronze Prize)</i> International Conference on Electronics, Information, and Communication (ICEIC) Title – LoRCoN-LO: Long-Term Recurrent Convolutional Network-Based LiDAR Odometry	Feb. 2023
<i>Outstanding Student Paper Award</i> Summer Annual Conference of the Institute of Electronics and Information Engineers (IEIE) Title – Multi-Floor Mapping including the Elevator	Jun. 2022

Work Experience

<i>Samsung Electronics</i> Future Robotics Office Staff Engineer Developing perceptive locomotion control algorithms for humanoid robots.	Seoul, Republic of Korea Mar. 2026 - Present
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Skills

Software and Programming Skills ROS 1/2 (Robot Operating System), C++, Python, PyTorch, OpenCV, g2o, Isaac Sim
Sensors LiDAR, Cameras, Sonar, IMUs, GPS, Encoders
Platforms Humanoid Robots, Quadruped Robots, Mobile Robots, Autonomous Vehicles

URLS

- Website (<https://donghwi jung.github.io>)
- Github (<https://github.com/donghwi jung>)
- Youtube (https://www.youtube.com/channel/UCqU4Gn7X3WlivtAI_yRaX0w)
- LinkedIn (<https://www.linkedin.com/in/donghwi-jung-6b9057a9/>)